

Lead Management System (LMS) - Backend Engineering Document

Purpose:

This document defines the complete backend architecture and engineering standards for building an enterprise-grade Lead Management System REST API.

Backend Framework:

Laravel 12

API Style:

REST API

Database:

MySQL

1. Backend Technology Stack

Framework:

Laravel 12

Database:

MySQL

Authentication:

Laravel Sanctum / JWT Authentication

Authorization:

Role Based Access Control (RBAC)

File Storage:

Local Storage / Cloud Storage Ready

API Communication:

JSON REST APIs

2. Laravel Project Architecture

Recommended Structure:

app/

Models/

Database entities

Http/Controllers/

API controllers

Services/

Business logic layer

Requests/
Validation classes

Middleware/
Authentication and permission checking

Resources/
API response formatting

Policies/
Authorization rules

Jobs/
Background processing

Notifications/
System alerts

3. Authentication Module

Features:

- User Login
- Token Generation
- Token Validation
- Logout
- Protected APIs
- User Session Management

Authentication Flow:

User Credentials
↓
Validate User
↓
Generate API Token
↓
Return User + Permissions
↓
Frontend stores token
↓
Protected API access

4. Role Based Access Control

Roles:

Super Admin

Admin

Sales Manager

Sales Executive

Marketing User

Viewer

Permission Examples:

create_lead

edit_lead

delete_lead

assign_lead

view_reports

manage_users

5. Core Backend Modules

Modules:

Authentication Module

User Management Module

Role Permission Module

Lead Management Module

Lead Assignment Module

Follow-up Module

Communication Module

Document Upload Module

Notification Module

Reports Module

Activity Log Module

6. API Endpoints

Authentication:

POST /api/login

POST /api/logout

Dashboard:

GET /api/dashboard

Leads:

GET /api/leads

POST /api/leads

GET /api/leads/{id}

PUT /api/leads/{id}

DELETE /api/leads/{id}

Follow-ups:

GET /api/followups

POST /api/followups

Users:

GET /api/users

POST /api/users

PUT /api/users/{id}

Reports:

GET /api/reports

7. Lead Management Business Logic

Backend responsibilities:

- Validate lead data
- Create leads

- Update lead information
- Track status changes
- Assign users
- Store notes
- Upload documents
- Maintain timeline
- Generate activity logs

8. Lead Assignment Logic

Supported Assignment Types:

Manual Assignment

Round Robin Assignment

Department Based Assignment

Rules:

Every lead should have ownership.

Assignment changes should create activity logs.

9. Notification System

Notification Events:

New lead assigned

Follow-up reminder

Missed follow-up

Status updated

Notification Storage:

Database notifications table

10. Reporting Module

Reports:

Daily Lead Reports

Weekly Reports

Monthly Reports

Sales Executive Performance

Lead Source Analysis

Conversion Reports

Revenue Reports

11. API Response Standard

Every API should follow a common format:

```
{  
  success:true,  
  message:"",  
  data:{}  
}
```

Errors:

```
{  
  success:false,  
  message:"",  
  errors:{}  
}
```

12. Security Requirements

Production Security:

- API Authentication
- Middleware Protection
- Input Validation
- SQL Injection Protection
- Rate Limiting
- Permission Checking
- Secure File Uploads
- Error Handling

13. Production Development Rules

Backend must follow:

Clean Architecture

Controller-Service Pattern

Reusable Code

Database Optimization

Proper Relationships

Error Handling

Scalable API Design

Maintainable Code Standards